MARKING SCHEME

Model Paper — 06

Third Term Test

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Note (for teachers):

- Part I – A: 2 marks per question (normally 1 mark for the method and 1 mark for the final answer; where there is no working, 1 mark for the answer and 1 mark for correct form/units). — Total 30 marks (15 questions).
- Part I – B: 10 marks per question (5 parts × 2 marks). — Total 20 marks. Part II: candidates answer any five questions, 10 marks each. — Total 50 marks. Step marks are shown in the right-hand column.
- Award full marks for correct alternative methods. Accept reasonable ranges for values read from graphs or measured from drawings. Follow-through marks may be given for correct working with an earlier wrong value.

PART I – A — Answers

1. Pattern: $1^2, 2^2, 3^2, 4^2, \ldots$; $T_n = n^2$; $T_{12} = 12^2$

Final answer: General term n^2 ; 12th term = 144

(2 marks)

2. Largest 4-digit binary number = 1111_2 ; $1\times8 + 1\times4 + 1\times2 + 1\times1$

Final answer: $1111_2 = 15$ (base ten)

(2 marks)

3. $\frac{1}{4} \times \frac{2}{3} = \frac{1}{6}$; $\frac{5}{6} - \frac{1}{6} = \frac{4}{6}$

Final answer: $\frac{2}{3}$

(2 marks)

4. Reduction = 800 – 680 = Rs 120 ; $\frac{120}{800} \times 100$

Final answer: 15%

(2 marks)

5. Two numbers with product –15 and sum –2: –5 and +3 ; $a^2 - 5a + 3a - 15 = a(a - 5) + 3(a - 5)$

Final answer: $(a - 5)(a + 3)$

(2 marks)

6. $\frac{1}{8} = \frac{1}{2^3}$

Final answer: 2^{-3}

(2 marks)

7. 6.05×10^{-3} : move the decimal point 3 places to the left

Final answer: 0.00605

(2 marks)

8. $0.5x = 8 - 3 = 5$; $x = 5 \div 0.5$

Final answer: x = 10

(2 marks)

9. Both sides of $2a = 14$ are divided by 2 (a non-zero number)

Final answer: If equals are divided by the same non-zero number, the results are equal.

(2 marks)

10. Angles on a straight line: $2x + 3x + 4x = 180°$; $9x = 180°$

Final answer: x = 20

(2 marks)

11. Perimeter = $2r + \frac{1}{4} \times 2\pi r = 28 + \frac{1}{4} \times 2 \times \frac{22}{7} \times 14 = 28 + 22$

Final answer: 50 cm

(2 marks)

12. $b^2 = 13^2 - 12^2 = 169 - 144 = 25$ (Pythagoras)

Final answer: 5 cm

(2 marks)

13. $\frac{x}{3} \geq 2 + 1 = 3$; $x \geq 9$

Final answer: $x \geq 9$

(2 marks)

14. Pick 2 of the 3 elements

Final answer: {a, b}, {a, c}, {b, c}

(2 marks)

15. Exterior angle = $180° - 160° = 20°$; $n = 360° \div 20°$

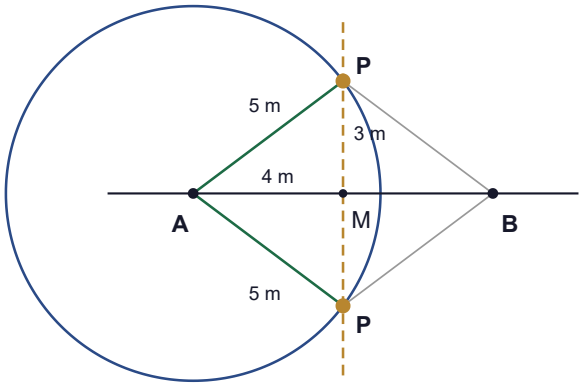
Final answer: 18 sides

(2 marks)

PART I – B — Answers and Mark Allocation

Question 1

(i)	
Circle with centre A and radius 5 m	**2**
(ii)	
The perpendicular bisector of AB (the straight line through the mid-point of AB, perpendicular to AB)	**2**
(iii)	
Sketch: line AB with the circle (centre A, radius 5 m) and the perpendicular bisector of AB (4 m from A) drawn	**1**
The two points where the circle meets the bisector marked P (one on each side of AB), each 3 m from the line AB	**1**



Rough sketch for Part I – B Question 1 (iii) (not to scale)

(iv)	
M = mid-point of AB, so AM = 4 m ; in the right-angled triangle AMP: $AP^2 = AM^2 + PM^2 \Rightarrow 25 = 16 + PM^2$	**1**
$PM^2 = 9 \Rightarrow PM = 3$ m (distance from P to AB)	**1**
(v)	
Area $= \frac{1}{2} \times AB \times PM = \frac{1}{2} \times 8 \times 3$	**1**
$= 12$ m^2	**1**

Total marks: 10

Question 2

(i)	
Volume = $40 \times 25 \times 30$	**1**
= 30 000 cm^3	**1**
(ii)	
1 litre = 1000 cm^3 ; 30 000 ÷ 1000	**1**
= 30 litres	**1**
(iii)	
30 litres = 30 000 ml ; 30 000 ÷ 250	**1**
= 120 cups	**1**
(iv)	
Cost = 30×120	**1**
= Rs 3 600	**1**
(v)	
Income = 120×40 = Rs 4 800 ; profit = 4 800 – 3 600 = Rs 1 200	**1**
Profit % = $\frac{1200}{3600} \times 100 = 33\frac{1}{3}\%$ ($\approx 33.3\%$)	**1**

Total marks: 10

PART II — Answers and Mark Allocation (any five questions)

Question 1

(a)	
Salary before increase = 100%, after = 112%	**1**
112% = 56 000 ⇒ 100% = $\frac{56000}{112} \times 100$	**1**
= Rs 50 000	**1**
(b)	
Selling price = 115% of cost price	**1**
115% = 1380 ⇒ 100% = $\frac{1380}{115} \times 100$	**1**
= Rs 1 200	**1**
(c)	
Girls = 100% – 40% = 60%	**1**
60% = 360 ⇒ total = $\frac{360}{60} \times 100$	**1**
Total = 600 students	**1**
Boys = 40% of 600 = 240 (or 600 – 360)	**1**

Total marks: 10

Question 2

(a)	
(i) $25 - x^2 = 5^2 - x^2$	**1**
$= (5 - x)(5 + x)$	**1**
(ii) $3x^2 - 12x = 3x(x - 4)$	**1**
(iii) $x^2 + 10x + 25 = x^2 + 2(x)(5) + 5^2$	**1**
$= (x + 5)^2$	**1**
(b)	
Common denominator 6: $\frac{2(2x + 1) - (x - 2)}{6}$	**1**
$= \frac{4x + 2 - x + 2}{6}$	**1**
$= \frac{3x + 4}{6}$	**1**
(c)	
$x^2 - y^2 = (x + y)(x - y)$	**1**
$= 7 \times 3 = 21$	**1**

Total marks: 10

Question 3

(i)	
Suitable scale and labelled axes (x: time in hours, y: length in cm)	**1**
Points (0, 20), (1, 17), (2, 14), (3, 11), (4, 8) all plotted correctly	**1**
A straight line through the points (decreasing, starting at (0, 20))	**1**
(ii)	
Gradient = $\frac{17-20}{1-0} = -3$ (or $\frac{8-20}{4} = -3$)	**1**
Meaning: the candle burns down (its length decreases) by 3 cm every hour	**1**
(iii)	
m = –3, intercept c = 20 (value of y when x = 0)	**1**
y = –3x + 20	**1**
(iv)	
x = 2.5 : y = –3(2.5) + 20 = 12.5 cm (reading from the graph accepted: 12.5 cm)	**1**
(v)	
Burns out when y = 0 : 0 = –3x + 20	**1**
$x = \frac{20}{3} = 6\frac{2}{3}$ hours (6 hours 40 minutes)	**1**

Total marks: 10

Question 4

(a)	
Angle sum of a triangle: (x + 10) + (2x – 20) + (x + 30) = 180	**1**
4x + 20 = 180 ⇒ x = 40	**1**
Angles: 50°, 60°, 70° (sum 180°)	**1**
(b)	
Sugar Rs s per kg, dhal Rs d per kg : 3s + 2d = 1440 .....①, 2s + 3d = 1610 .....②	**1**
①×3 : 9s + 6d = 4320 ; ②×2 : 4s + 6d = 3220	**1**
Subtracting: 5s = 1100	**1**
s = 220 (sugar Rs 220 per kg)	**1**
Substituting in ① : 660 + 2d = 1440 ⇒ d = 390 (dhal Rs 390 per kg) ; check in ② : 440 + 1170 = 1610 ✓	**1**
(c)	
$P = 2l + 2b \Rightarrow 2b = P - 2l \Rightarrow b = \frac{P - 2l}{2}$ (or $b = \frac{P}{2} - l$)	**1**
$b = \frac{50 - 30}{2} = 10$	**1**

Total marks: 10

Question 5

(a)	
$A\hat{B}C = A\hat{D}E = 55°$ (corresponding angles, DE ∥ BC)	**1**
$A\hat{C}B = A\hat{E}D = 65°$ (corresponding angles, DE ∥ BC)	**1**
$B\hat{A}C = 180° - (55° + 65°)$ (angle sum of triangle ABC)	**1**
$B\hat{A}C = 60°$	**1**
(b)	
Sum of interior angles = (6 – 2) × 180° = 720°	**1**
Each angle of a regular hexagon = 720° ÷ 6 = 120°	**1**
(c)	
Sum of exterior angles = 360° : x + 2x + 3x + 4x = 360°	**1**
10x = 360° ⇒ x = 36°	**1**
Largest interior angle belongs to the smallest exterior angle (x = 36°)	**1**
Largest interior angle = 180° – 36° = 144°	**1**

Total marks: 10

Question 6

(a)(i)	
Two straight parts = $2 \times 100 = 200$ m	**1**
Two semicircles = one circle of diameter 70 m : $\pi d = \frac{22}{7} \times 70 = 220$ m	**1**
Perimeter = 200 + 220 = 420 m	**1**
(a)(ii)	
Rectangle = $100 \times 70 = 7\,000$ m^2	**1**
Circle (radius 35 m) = $\frac{22}{7} \times 35 \times 35 = 3\,850$ m^2	**1**
Area = 7 000 + 3 850 = 10 850 m^2	**1**
(b)	
(i) Area = base × height $\Rightarrow 96 = 12 \times h$	**1**
$h = 96 \div 12 = 8$ cm	**1**
(ii) $96 = BC \times h' = 10 \times h'$	**1**
$h' = 96 \div 10 = 9.6$ cm	**1**

Total marks: 10

Question 7

(i)	
14 + 10 + 10 + 6 = 40 students	**1**
(ii)	
Mango (the tallest bar, 14)	**1**
(iii)	
Bananas = 10 students : $\frac{10}{40} \times 100$	**1**
= 25%	**1**
(iv)	
Mango = 14 students	**1**
Angle = $\frac{14}{40} \times 360°$	**1**
= 126°	**1**
(v)	
Papaya = 10 ; P = $\frac{10}{40}$	**1**
= $\frac{1}{4}$	**1**
(vi)	
14 – 6 = 8 students	**1**

Total marks: 10